Power BI Project Documentation

Customer Analytics Dashboard

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1. PROJECT OBJECTIVE

The objective of this project is to design and develop an interactive Customer Analytics Dashboard using Power BI to:

- Analyze customer behavior and engagement
- Track churn and retention patterns
- Evaluate revenue performance
- Enable data-driven decision-making
- Provide actionable insights to improve customer lifecycle management

2. PROJECT OVERVIEW

This project integrates multiple datasets including:

- Customer Master Data
- Transactions Data
- Subscription Data
- Engagement Data

3. OVERVIEW OF DASHBOARDS

The dashboard is structured into 5 analytical pages, each focusing on a specific business area:

1. Customer Overview

This page provides a consolidated view of key metrics such as total customers, revenue, and active users.

It enables quick assessment of overall business health and customer base performance.

2. Customer Segmentation

This page categorizes customers based on attributes like region, gender, and subscription type.

It helps in identifying different customer groups for targeted marketing and strategy planning.

3. Engagement Analysis

This page analyzes customer interaction through metrics such as login frequency and session duration.

It helps in evaluating user activity levels and identifying highly engaged users.

4. Churn & Retention Insights

This page focuses on tracking customer churn and retention rates over time.

It helps in understanding customer behavior and developing strategies to reduce churn.

5. Revenue & Performance Analysis

This page presents revenue trends, transaction patterns, and customer spending behavior.

It helps in assessing financial performance and identifying growth opportunities.

3. DATA MODEL & RELATIONSHIPS

Tables Used:

Table Name	Description
Customers	Customer Demographic Details
Transactions	Purchase Transactions
Subscriptions	Subscription status and renewal
Engagement	Customer activity and interaction
DateTable	Calendar table for time analysis

Relationship Structure:

From Table	Column	To Table	Column	Type
Customers	Customer ID	Transactions	Customer ID	1 : Many
Customers	Customer ID	Subscriptions	Customer ID	1 : Many
Customers	Customer ID	Engagement	Customer ID	1 : Many
Date Table	Date	Transactions	Purchase Date	1 : Many (Active)
Date Table	Date	Subscriptions	Renewal Date	1 : Many (Active)

Model Design Approach

Component	Description
Data Model Type	Star Schema was implemented for better performance and scalability
Dimension tables	Customers_Masters and dateTable act as dimension tables
Fact Tables	Transactions, Subscriptions and Engagement act as fact tables
Filtering method	Single-Direction filtering is used to avoid ambiguity
Relationship Strategy	Direct relationships between fact tables are avoided to maintain model clarity

4. DATA TRANSFORMATION

The following transformations were applied:

Data Cleaning

• Removed null and duplicate records

• Standardized column names

• Converted date columns to proper Date format

Derived Columns - Customers_Masters Table

Column Name	DAX Formula	Purpose
Age Group	`Age Group = SWITCH(TRUE(), Customer_Master[Age] < 25, "Young", Customer_Master[Age] < 45, "Middle", "Senior")`	Segments customers into age categories for analysis

Derived Columns - Subscription Table

Column Name	DAX Formula	Purpose
Churn Flag	`ChurnFlag = IF(Subscription[Status] <> "Active", 1, 0)`	Identifies Churned Customers
Is Active	`IsActive = IF(Subscription[Status] = "Active",1,0)`	Identifies Currently active customers

Derived Columns - Subscription Table

Column Name	DAX Formula	Purpose
Month	`Month = FORMAT(Transactions[PurchaseDate],"MMM")`	Enables monthly trend analysis
Year	`Year = YEAR(Transactions[PurchaseDate])`	Enables yearly aggregation
Customer Type	`Customer Type = IF(RELATED(Subscription[Status]) = "Active", "Subscribed", "Non-subscribed")`	Classifies customers based on transaction value

Data Formatting

• Ensured consistent data types across tables

• Applied proper sorting for Month-Year fields

Importance of Date Table

Aspect	Description
Purpose	A DateTable is created to enable efficient time-based analysis across all datasets
Time Intelligence	Supports calculations like monthly trends, yearly growth, and YTD analysis
Central Filtering	Acts as a common filter across Transactions, Subscriptions, and other fact tables
Relationship Hub	Connects with multiple tables using date fields (e.g., PurchaseDate, RenewalDate)
Consistency	Ensures uniform date hierarchy (Year, Month, Quarter) across all visuals

5. DAX MEASURES CREATED

Key Metrics

DAX Measures Customers_Masters Table

Measure Name	DAX Formula	Purpose
Total Customers	`TotalCustomers = DISTINCTCOUNT(Customer_Master[CustomerID])`	Calculates total unique customers
Total Revenue	`TotalRevenue = SUM(Transactions[Amount])`	Calculates overall revenue generated
Total Transactions	`TotalTransactions = COUNT(Transactions[TransactionID])`	Counts total number of transactions

DAX Measures Engagement Table

Measure Name	DAX Formula	Purpose
Avg Purchase Value	`Avg Purchase Value = DIVIDE([TotalRevenue], [TotalTransactions])`	Calculates average value per transaction
Avg Revenue per Customer	`Avg Revenue per Customer = DIVIDE([TotalRevenue], [TotalCustomers])`	Measures customer spending value
Avg Transaction per Customer	`Avg Transactions per Customer = DIVIDE([TotalTransactions],[TotalCustomers])`	Shows average transactions per customer

DAX Measures Subscription Table

Measure Name	DAX Formula	Purpose
Active Customers	`ActiveCustomers = CALCULATE(DISTINCTCOUNT(Customer_Master[CustomerID]), Subscription[Status] = "Active")`	Counts currently active users
Churned Customers Trend	`Churned Customers Trend = CALCULATE(DISTINCTCOUNT(Subscription[CustomerID]), Subscription[Status] <> "Active", USERELATIONSHIP(DateTable[Date], Subscription[RenewalDate]))`	Tracks churn over time
Retention rate	`Retention Rate = DIVIDE([ActiveCustomers], [TotalCustomers])`	Measures customer retention percentage
Revenue Growth	`RevenueGrowth = VAR Prev = CALCULATE([TotalRevenue], DATEADD(DateTable[Date],`	Calculates month-overtime revenue growth

	`-1, MONTH)) RETURN DIVIDE([TotalRevenue] - Prev, Prev)`	
Churn Rate	`ChurnRate = DIVIDE([ChurnedCustomers], [TotalCustomers])`	Calculates the percentage of customers who have churned

DAX Measures Transaction Table

Measure Name	DAX Formula	Purpose
Monthly Active Customers	`Monthly Active Customers = CALCULATE(DISTINCTCOUNT(Transactions[CustomerID]))`	Tracks active customers per month

6. DASHBOARD PAGES DESCRIPTION

PAGE 1: Customer Overview

KPI Cards

• Total Customers → [Total Customers]

• Total Revenue → [Total Revenue]

• Active Customers → [Active Customers]

• Avg Revenue per Customer → [Avg Revenue per Customer]

1. Line Chart (Revenue Trend)

• X-axis: DateTable[YearMonth]

• Y-axis (Values): [Total Revenue]

• Legend: None

• Tooltips: [Total Transactions]

2. Bar Chart (Customers by Region)

• X-axis: Customers[Region]

• Y-axis: [Total Customers]

3. Donut Chart (Gender Distribution)

• Legend: Customers[Gender]

• Values: [Total Customers]

4. Column Chart (Age Group)

• X-axis: Customers[Age Group]

• Y-axis: [Total Customers]

PAGE 2: Customer Segmentation

KPI Cards

• Total Customers → [Total Customers]

• Subscribed Customers → [Active Customers]

• Non-Subscribed → [Churned Customers]

• Avg Transactions per Customer → [Avg Transactions per Customer]

1. Stacked Bar (Region vs Customer Type)

• X-axis: Customers[Region]

• Y-axis: [Total Customers]

• Legend: Transactions[Customer Type]

2. Treemap (Product Revenue)

• Group: Transactions[Product]

• Values: [Total Revenue]

3. Clustered Column (Age vs Revenue)

• X-axis: Customers[Age Group]

• Y-axis: [Total Revenue]

4. Donut Chart (Plan Type)

• Legend: Subscriptions[PlanType]

• Values: [Total Customers]

PAGE 3: Engagement Analysis

KPI Cards

• Avg Transactions per Customer → [Avg Transactions per Customer]

• Avg Purchase Value → [Avg Purchase Value]

• Monthly Active Customers → Monthly Active Customers

1. Line Chart (Transactions Trend)

• X-axis: DateTable[YearMonth]

• Y-axis: [Total Transactions]

2. Matrix (Heatmap Style)

• Rows: DateTable[Month]

• Columns: Customers[Region]

• Values: [Total Transactions]

3. Scatter Plot (Customer Behavior)

• X-axis: [Total Transactions]

• Y-axis: [Total Revenue]

• Size: [Total Customers]

• Legend: Customers[Region]

• Details: Customers[CustomerID]

4. Bar Chart (Top Products)

• X-axis: Transactions[Product]

• Y-axis: [Total Transactions]

• Legend: None

• Filter: Top N = 10

PAGE 4: Churn & Retention Insights

KPI Cards

• Churned Customers → [Churned Customers]

• Churn Rate → [Churn Rate]

• Retention Rate → [Retention Rate]

• Active Customers → [Active Customers]

1. Line Chart (Churn Trend)

• X-axis: DateTable[YearMonth]

• Y-axis: [Churned Customers]

2. Bar Chart (Churn by Region)

• X-axis: Customers[Region]

• Y-axis: [Churned Customers]

3. Donut Chart (Active vs Churned)

• Legend: Subscriptions[Status]

• Values: [Total Customers]

4. Column Chart (Plan vs Churn)

• X-axis: Subscriptions[PlanType]

• Y-axis: [Churn Rate]

PAGE 5: Revenue & Performance Analysis

KPI Cards

• Total Revenue → [Total Revenue]

• Revenue Growth → [Revenue Growth]

• Avg Revenue per Customer → [Avg Revenue per Customer]

• Total Transactions → [Total Transactions]

1. Line Chart (Revenue Trend)

• X-axis: DateTable[YearMonth]

• Y-axis: [Total Revenue]

2. Bar Chart (Revenue by Product)

• X-axis: Transactions[Product]

• Y-axis: [Total Revenue]

3. Stacked Column (Region + Plan Revenue)

• X-axis: Customers[Region]

• Y-axis: [Total Revenue]

• Legend: Subscriptions[PlanType]

4. Waterfall Chart (Revenue Contribution)

• Category: Transactions[Product]

• Y-axis: [Total Revenue]

Slicers & Drill-Through

• DateTable[Year]

• Customers[Region]

• Subscriptions[PlanType]

• Subscriptions[Status]

➔ Drill Through is added between customer segmentation and Revenue & performance analysis for deep analysis

7. PROJECT OUTCOMES

• Developed a fully interactive dashboard for business insights

• Enabled real-time tracking of KPIs

• Improved visibility into customer lifecycle and churn behavior

• Provided actionable insights for:

• Marketing strategies

• Customer retention

• Revenue optimization

8. CONCLUSION

The Customer Analytics Dashboard successfully transforms raw data into meaningful insights. It empowers decision-makers with:

• Clear visualization of trends

• Improved customer understanding

• Data-driven strategic planning

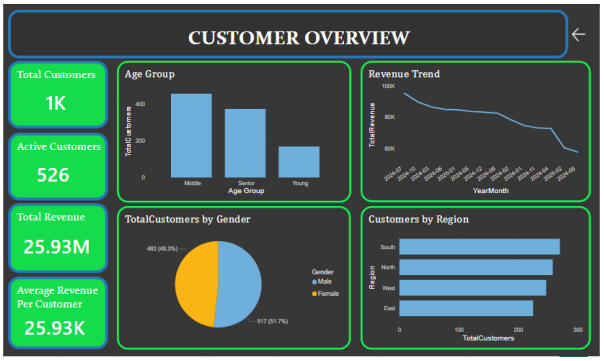
9. FUTURE ENHANCEMENTS

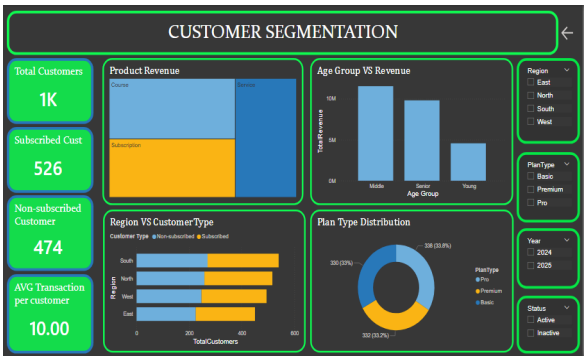
• Cohort-based churn analysis

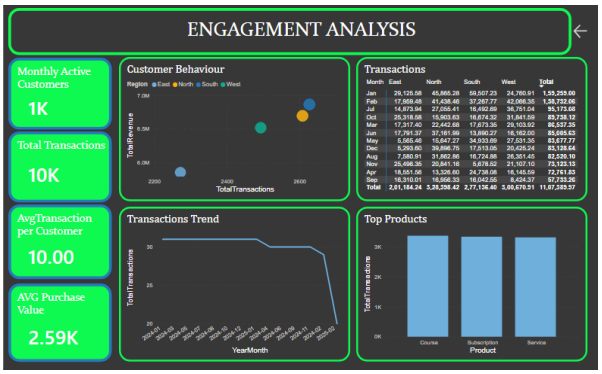
• Customer Lifetime Value (LTV) modeling

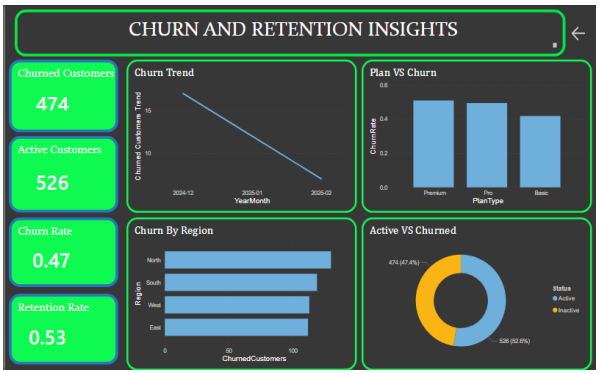
• Predictive analytics using machine learning

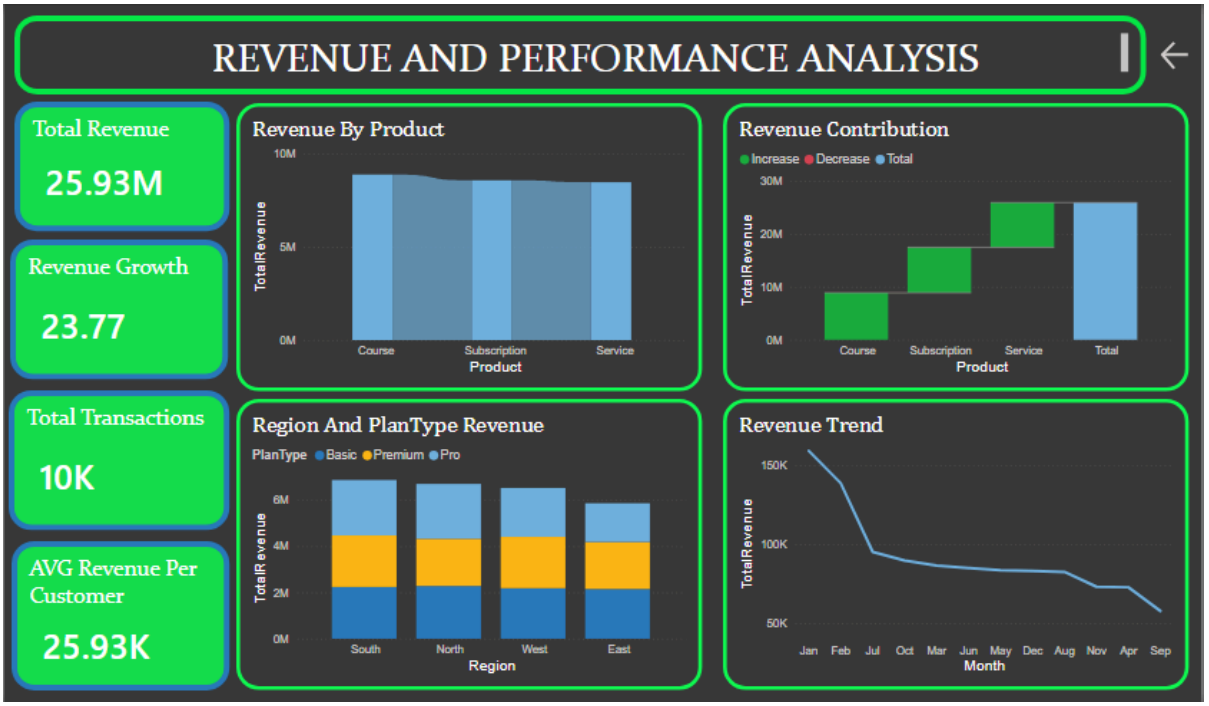
• Real-time data integration


CUSTOMER OVERVIEW
Total Customers
1K
Active Customers
526
Total Revenue
25.93M
Average Revenue Per Customer
25.93K
Age Group
TotalCustomers
Middle
Senior
Young
Age Group
Revenue Trend
TotalRevenue
YearMonth
TotalCustomers by Gender
483 (48.3%)
517 (51.7%)
Gender
Male
Female
Customers by Region
Region
South
North
West
East
TotalCustomers


CUSTOMER SEGMENTATION
Total Customers
1K
Subscribed Cust
526
Non-subscribed Customer
474
AVG Transaction per customer
10.00
Product Revenue
Course
Service
Subscription
Age Group VS Revenue
TotalRevenue
Middle
Senior
Young
Age Group
Region
East
North
South
West
PlanType
Basic
Premium
Pro
Year
2024
2025
Status
Active
Inactive
Region VS CustomerType
Customer Type
Non-subscribed
Subscribed
Region
South
North
West
East
TotalCustomers
Plan Type Distribution
330 (32%)
338 (33.8%)
332 (33.2%)
PlanType
Pro
Premium
Basic


ENGAGEMENT ANALYSIS
Monthly Active Customers
1K
Total Transactions
10K
AvgTransaction per Customer
10.00
AVG Purchase Value
2.59K
Customer Behaviour
Region
East
North
South
West
TotalRevenue
TotalTransactions
Transactions
Transactions Trend
TotalTransactions
YearMonth
Top Products
TotalTransactions
Course
Subscription
Service
Product


CHURN AND RETENTION INSIGHTS
Churned Customers
474
Active Customers
526
Churn Rate
0.47
Retention Rate
0.53
Churn Trend
Churned Customers Trend
2024-12
2025-01
2025-02
YearMonth
Plan VS Churn
ChurnRate
Premium
Pro
Basic
PlanType
Churn By Region
Region
North
South
West
East
ChurnedCustomers
Active VS Churned
474 (47.4%)
526 (52.6%)
Status
Active
Inactive


REVENUE AND PERFORMANCE ANALYSIS
Total Revenue
25.93M
Revenue Growth
23.77
Total Transactions
10K
AVG Revenue Per Customer
25.93K
Revenue By Product
10M
5M
0M
TotalRevenue
Course
Subscription
Service
Product
Revenue Contribution
Increase
Decrease
Total
30M
20M
10M
0M
TotalRevenue
Course
Subscription
Service
Total
Product
Region And PlanType Revenue
PlanType
Basic
Premium
Pro
6M
4M
2M
0M
TotalRevenue
South
North
West
East
Region
Revenue Trend
150K
100K
50K
TotalRevenue
Jan
Feb
Jul
Oct
Mar
Jun
May
Dec
Aug
Nov
Apr
Sep
Month